1. Introduction

Interlock is a database system for multi-team mechanical assembly data. The identity of a part is computed from its geometry, the relationships between parts are declared as machine-checkable interface contracts, and every proposed change is validated against those contracts before it is committed. It ingests assemblies from existing CAD tools and supports **SDG 9 (Industry, Innovation and Infrastructure)** by reducing integration rework, duplicate parts and late-stage redesign.

2. Problem Statement

When several teams build one machine, their parts meet for the first time at integration. A mounting hole moves two millimetres and nobody downstream is told; two parts are each within tolerance but their combined error is not; the same bracket exists under six part numbers. CAD tools model shape but not ownership or dependency, while PLM tools track part numbers but treat geometry as an opaque attachment. The gap between them is filled today by spreadsheets and manual review.

3. Literature Review

[1] Osada, R., Funkhouser, T., Chazelle, B., and Dobkin, D., "Shape Distributions," ACM Transactions on Graphics, 21(4), 2002. A solid can be reduced to a compact signature that is invariant under translation and rotation, so shape comparison becomes a comparison of fixed-length descriptors. The signature is stable across tessellation and modelling history, which allows the same part exported by two systems to be recognised as one.

[2] Tangelder, J. W. H., and Veltkamp, R. C., "A survey of content based 3D shape retrieval methods," Multimedia Tools and Applications, 39(3), 2008. Descriptors built from global properties (volume, area, moments of inertia, surface-type statistics) are far cheaper to compute and index than graph-based descriptors yet sufficient for exact matching; the quantisation precision is a primary design parameter.

[3] Babic, B., Nesic, N., and Miljkovic, Z., "A review of automated feature recognition with rule-based pattern recognition," Computers in Industry, 59(4), 2008. Cylindrical and planar faces carry most of a part's assembly intent, making hole patterns and mounting planes the natural units of an interface. Automatic recovery of intent remains unreliable, so interfaces should be declared by the designer and verified against geometry.

[4] Merkle, R. C., "A Digital Signature Based on a Conventional Encryption Function," CRYPTO '87, LNCS 293, Springer, 1988. A hash tree summarises a large structure in one root value, so equality of two structures is a single comparison, and a local change alters only the hashes on its path to the root.

[5] ISO 10303-242:2020, Managed model-based 3D engineering (STEP AP242), ISO, Geneva, 2020. The neutral exchange format carries the assembly hierarchy and placement transforms natively, but it records where a part is, not why; mating relationships must therefore be authored explicitly.

[6] Jarratt, T. A. W., Eckert, C. M., Caldwell, N. H. M., and Clarkson, P. J., "Engineering change: an overview and perspective on the literature," Research in Engineering Design, 22(2), 2011. Changes propagate along component interfaces and their cost rises sharply with the stage at which they are found; impact is routinely underestimated because no one holds the full dependency graph.

[7] Chase, K. W., and Parkinson, A. R., "A survey of research in the application of tolerance analysis to the design of mechanical assemblies," Research in Engineering Design, 3(1), 1991. Tolerances accumulate along an assembly path, so individually conforming

parts can form a non-conforming assembly; worst-case and statistical (RSS) stacking trade cost against yield, so both totals should be kept.

Research gap: each of these ideas is well established on its own, but none of the works combines them in a single store in which the shape descriptor is the primary key, the assembly hash is built from those keys, and interface and tolerance constraints are checked when a change is committed.

4. Existing Systems

Table 4.1: Comparison with existing tools

Category	Examples	Unit of identity	Limitation
PDM / file vaults	SolidWorks PDM, Autodesk Vault	File and assigned part number	Duplicates go undetected; review is manual
PLM suites	Teamcenter, Windchill, ENOVIA	Human-assigned part number	Geometry is an opaque attachment
Cloud CAD	Onshape	Document and element ID	Tied to one authoring tool; no cross-team contracts
Version control	Git with LFS	Hash of file bytes	A re-save with no edit looks like a change
Shape search	Siemens Geolus, CADENAS	Shape descriptor, for ranking	Advisory only; cannot deduplicate on write
Tolerance analysis	CETOL 6σ, 3DCS	Per-analysis study	Run offline, detached from the data store
Interlock (proposed)	**—**	**Hash of a pose-invariant geometric vector**	**Contracts and tolerances checked at commit**

The closest existing work is shape-similarity search, which shows that descriptor-based comparison is industrially viable but uses the descriptor only to rank candidates for reuse; it is not the key of the store, so it cannot prevent duplicates or suppress a spurious change. Tolerance analysis suites evaluate stacks accurately, but each study is maintained beside the design data and re-run only when an analyst remembers to, rather than being enforced on every change.

Part numbers miss every geometric change and byte hashes flag every re-save; a geometric fingerprint ignores re-save, rename and rotation yet detects a hole widened by 0.2 mm (mirror-image parts need one extra signed quantity).

5. Technology Stack

Python 3.10; SQLite with recursive queries; Open CASCADE geometry kernel (cadquery-ocp), used headlessly and only at import; NumPy and SciPy; TinyDB; FastAPI with Jinja2 for the review interface; pytest for testing.

6. Expected Outcome

- Parts identified by geometry, so independent exports of one assembly create no duplicates, and re-saves are no-ops.
- Integration errors caught at proposal time, with the violated interface, tolerance chain and affected team named.
- Lock-free concurrent work, with each full build configuration identified by a single root hash.